```html
<!doctype html>
<html lang="en">
  <head>
    <!-- Required meta tags -->
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">

    <!-- Bootstrap CSS -->
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.0.2/dist/css/bootstrap.min.css" rel="stylesheet" integrity="sha384-EVSTQN3/azprG1Anm3QDgpJLIm9Nao0Yz1ztcQTwFspd3yD65VohhpuuCOmLASjC" crossorigin="anonymous">

    <title>Game Simulator</title>
  <style>
      body {
        background-color: #f5f5f5;
        color: #333;
        font-family: "Segoe UI", Arial, sans-serif;
      }
      h1, h2, h3 {
        color: #2e7d32; /* green accent */
      }
      .navbar {
        background-color: #263238 !important; /* deep navy */
      }
      .navbar-brand, .navbar-nav .nav-link {
        color: #ffffff !important;
      }
      .section {
        margin-top: 2rem;
        margin-bottom: 2rem;
        padding: 1rem 1.5rem;
        background: #ffffff;
        border-radius: 10px;
        box-shadow: 0px 2px 6px rgba(0,0,0,0.1);
      }
      .bio-img {
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        max-width: 250px;
        border-radius: 50%;
        margin-bottom: 1.5rem;
        border: 4px solid #2e7d32;
      }
      hr {
        border-top: 2px solid #2e7d32;
        opacity: 0.6;
      }
      a {
        color: #0288d1;
        text-decoration: none;
      }
      a:hover {
        text-decoration: underline;
      }
    </style>
  </head>
  <body>
    <nav class="navbar navbar-expand-md">
      <a class="navbar-brand" href="#">BGF Analytics</a>
      <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#navbarNav">
        <span class="navbar-toggler-icon"></span>
      </button>
      <div class="collapse navbar-collapse" id="navbarNav">
        <ul class="navbar-nav ms-auto">
          <li class="nav-item"><a class="nav-link" href="{{ url_for('home_page')}}">Home Page</a></li>
          <li class="nav-item"><a class="nav-link" href="{{ url_for('resume')}}">Resume</a></li>
          <li class="nav-item"><a class="nav-link" href="{{ url_for('projects')}}">Projects</a></li>
          <li class="nav-item"><a class="nav-link" href="{{ url_for('game_sim')}}">Game Simulator</a></li>
          <li class="nav-item"><a class="nav-link" href="{{ url_for('XGB_sim')}}">Bracket Simulator</a
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></li>
        </ul>
      </div>
    </nav>
    <h1>March Madness Game Simulator</h1>
        <p>
            Welcome to the March Madness Game Simulator!
        </p>
        <p>
            Select the two teams, whose game you want to simulate.
        </p>
    <div class="container mt-4">
    <form id="simulationForm" method="POST" onsubmit="return false;">

     <!-- User Team Selector -->
    <div class="row mb-3">
      <div class="col-md-6">
        <label for="team1" class="form-label">Team 1</label>
        <select class="form-select" id="team1" name="team1" required>
          <option value="" disabled selected>Choose a team...</option>
          {% set teams = [
            "Auburn","Alabama State","Louisville","Creighton","Michigan",
            "UC San Diego","Texas A&M","Yale","Mississippi","North Carolina",
            "Iowa State","Lipscomb","Marquette","New Mexico","Michigan State",
            "Bryant","Florida","Norfolk State","Connecticut","Oklahoma",
            "Memphis","Colorado State","Maryland","Grand Canyon","Missouri",
            "Drake","Texas Tech","UNC Wilmington","Kansas","Arkansas",
            "St. John's","Nebraska Omaha","Duke","Mount St. Mary's",
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            "Mississippi State","Baylor","Oregon","Liberty","Arizona","Akron",
            "BYU","VCU","Wisconsin","Montana","Saint Mary's","Vanderbilt",
            "Alabama","Robert Morris","Houston","SIU Edwardsville","Gonzaga",
            "Georgia","Clemson","McNeese State","Purdue","High Point",
            "Illinois","Xavier","Kentucky","Troy","UCLA","Utah State",
            "Tennessee","Wofford"
          ] %}
          {% for team in teams %}
            <option value="{{ team }}">{{ team }}</option>
          {% endfor %}
        </select>
      </div>
      <div class="col-md-6">
        <label for="team2" class="form-label">Team 2</label>
        <select class="form-select" id="team2" name="team2" required>
          <option value="" disabled selected>Choose a team...</option>
          {% for team in teams %}
            <option value="{{ team }}">{{ team }}</option>
          {% endfor %}
        </select>
      </div>
    </div>

     <!-- User Feature Controls Description -->
        <div class="card bg-dark text-white mb-4">
          <div class="card-body">
            <h5 class="card-title">Simulation Controls Explained</h5>

            <div class="row">
              <div class="col-md-6 mb-3">
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                <p><strong>Chaos Level (Noise Std):</strong> Controls randomness in score predictions.
                  <br>• <em>Low (0.5)</em>: Very stable outcomes, few upsets.
                  <br>• <em>Medium (2.5)</em>: Balanced realism and unpredictability.
                  <br>• <em>High (5.0)</em>: Wild swings, frequent upsets and score volatility.
                </p>
              </div>

              <div class="col-md-6 mb-3">
                <p><strong>Seed Consistency Bias:</strong> Adjusts how much top seeds resist randomness.
                  <br>• <em>Low (0.5)</em>: Top seeds dominate, fewer surprises.
                  <br>• <em>Medium (0.8)</em>: Realistic balance of consistency and chaos.
                  <br>• <em>High (1.2)</em>: All seeds treated more equally, more parity.
                </p>
              </div>

              <div class="col-md-6 mb-3">
                <p><strong>Model vs History:</strong> Blends predictive model with historical seed win rates.
                  <br>• <em>Low (0.0)</em>: Pure model logic, no historical bias.
                  <br>• <em>Medium (0.5)</em>: Balanced between model and history.
                  <br>• <em>High (1.0)</em>: Pure historical outcomes, less model influence.
                </p>
              </div>

              <div class="col-md-6 mb-3">
                <p><strong>Normalize Scores by Round:</strong> Aligns scores with historical round averages.
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                    <br>• <em>Yes</em>: Scores reflect typical round scoring (e.g. lower in Final Four).
                    <br>• <em>No</em>: Raw model outputs used, may be inflated or deflated.
                  </p>
                </div>
              </div>
            </div>
          </div>

      <!-- User Simulation Varible Contols -->
      <div class="row mb-3">
         <div class="col-md-3">
            <label for="noise_std" class="form-label">Chaos Level (Noise Std)</label>
                  <select class="form-select" id="noise_std" name="noise_std">
                   {% for val in [0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0] %}
                      <option value="{{ val }}">{{ val }}</option>
                      {% endfor %}
                  </select>
      </div>

      <div class="col-md-3">
        <label for="consistency_multiplier" class="form-label">Seed Consistency Bias</label>
            <select class="form-select" id="consistency_multiplier">
              {% for val in [0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.1, 1.2] %}
                <option value="{{ val }}">{{ val }}</option>
              {% endfor %}
            </select>
      </div>

      <div class="col-md-3">
        <label for="blend_weight" class="form-label"
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>Model vs History</label>
        <select class="form-select" id="blend_weight" name="blend_weight">
          {% for val in [0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0] %}
            <option value="{{ val }}">{{ val }}</option>
          {% endfor %}
        </select>
      </div>

      <div class="col-md-3">
        <label for="apply_normalization" class="form-label">Normalize Scores by Round</label>
        <select class="form-select" id="apply_normalization" name="apply_normalization">
          <option value="True">Yes</option>
          <option value="False" selected>No</option>
        </select>
      </div>
    </div>
     <!-- Run Simulation Button -->
    <button id="simulateBtn" type="submit" class="btn btn-primary mt-3">Simulate Game</button>
  </form>

<!-- Fetch User Inputs -->
<script>
document.addEventListener("DOMContentLoaded", () => {
  console.log("□ Game Simulator script loaded");

  const btn = document.getElementById('simulateBtn');
  console.log("□ Button element is", btn);

  btn.addEventListener('click', function(e) {
    e.preventDefault();
    console.log("□ Simulate button clicked");

    const team1 = document.getElementById('team1').
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value;
    const team2 = document.getElementById('team2').value;

    if (!team1 || !team2) {
      alert('Please select both teams before simulating.');
      return;
    }

    const formData = new FormData(document.getElementById('simulationForm'));

    fetch('/simulate', { method: 'POST', body: formData })
      .then(r => r.json())
      .then(data => {
        if (data.error) {
          document.getElementById('resultArea').textContent = data.error;
        } else {
          document.getElementById('resultArea').innerHTML =
            `<div>${data.XGB_result}</div>`;
        }
      })
      .catch(err => {
        console.error("Fetch failed:", err);
        document.getElementById('resultArea').textContent =
          'Simulation failed. See console for details.';
      });
  });
});
</script>



<!-- Results Area -->
    <div id="resultArea" style="margin-top: 1em;
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font-weight: bold;"></div>
</div>
    <!-- Optional JavaScript; choose one of the two ! -->

    <!-- Option 1: Bootstrap Bundle with Popper -->
    <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.0.2/dist/js/bootstrap.bundle.min.js" integrity="sha384-MrcW6ZMFYlzcLA8Nl+NtUVF0sA7MsXsP1UyJoMp4YLEuNSfAP+JcXn/tWtIaxVXM" crossorigin="anonymous"></script>

    <!-- Option 2: Separate Popper and Bootstrap JS -->
    <!--
    <script src="https://cdn.jsdelivr.net/npm/@popperjs/core@2.9.2/dist/umd/popper.min.js" integrity="sha384-IQsoLXl5PILFhosVNubq5LC7Qb9DXgDA9i+tQ8Zj3iwWAwPtgFTxbJ8NT4GN1R8p" crossorigin="anonymous"></script>
    <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.0.2/dist/js/bootstrap.min.js" integrity="sha384-cVKIPhGWiC2Al4u+LWgxfKTRIcfu0JTxR+EQDz/bgldoEyl4H0zUF0QKbrJ0EcQF" crossorigin="anonymous"></script>
    -->
  </body>

</html>
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